

MALLIKARJUN NEERALAGI

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PROFILE SUMMARY

Electronics and Communication Engineering student with hands-on experience in Embedded Systems, PCB Design, and IoT Development. Proficient in Altium Designer, KiCad, ESP32, STM32, and Raspberry Pi, with practical exposure to SMT assembly and hardware testing. Seeking an internship to apply embedded systems and PCB design skills to real-world product development.

TECHNICAL SKILLS

- Programming Languages: C, C++, Embedded C, Python
- Embedded Platforms: Arduino, ESP32, STM32, Raspberry Pi
- PCB Design: Altium Designer, KiCad
- Tools & Software: STM32CubeIDE, Arduino IDE, Git, Matplotlib
- Hardware & Protocols: Sensors, WaveShare Displays, UART, SPI, I2C
- Manufacturing Exposure: SMT Assembly, Pick-and-Place Machines, PCB Inspection

PROJECTS

IoT-Based Home Automation System

- Developed an IoT-based home automation system using ESP32, relay modules, and Wi-Fi communication for remote appliance control.
- Improved energy efficiency through automated appliance scheduling and control.

ThermaGuard AI: Metal and Mobile Device Detection

- Designed a thermal detection system using an MLX90640 sensor to identify metal objects and mobile devices.
- Integrated thermal imaging with embedded hardware to enable real-time monitoring and alerts.

Privacy-First Healthcare Monitor

- Developed a healthcare monitoring system with local, on-device sensor-based data processing to preserve user privacy.

INDUSTRIAL TRAINING

Embedded Systems and PCB Design Trainee

Vithsutra Technologies Pvt. Ltd.

- Designed PCB layouts using Altium Designer and KiCad for embedded hardware projects.
- Observed and applied industry-standard PCB assembly and quality assurance workflows.
- Gained hands-on experience in SMT manufacturing and Pick-and-Place machine operation.
- Assisted in hardware testing, debugging, and system integration.

EDUCATION

Alva's Institute of Engineering and Technology, Moodbidri

2023 - 2027 (Expected)

Bachelor of Engineering in Electronics and Communication Engineering | CGPA: 8.4

Prerana PU College, Belagavi

2021 - 2023

Pre-University (PCMC) | Percentage: 86%

SRMRS, Khanapur

2021

SSLC | Percentage: 93%

CERTIFICATIONS

- NPTEL - Embedded Systems
- Scalar - Python Programming
- Coursera - Altium Design, C, C++, Embedded C